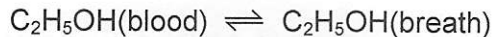


24 Ethanol in blood can be assumed to be in dynamic equilibrium with ethanol in breath.

This equilibrium forms the basis of the roadside breathalyser test.



The equilibrium constant, K_c , for this equilibrium (at body temperature and pressure) is 4.3×10^{-4} .

The concentration of ethanol in a motorist's blood is found to be 75 mg in 100 cm^3 of blood.

What is the concentration of ethanol in the motorist's breath in mg per 100 cm^3 ?

- A** 3.23×10^{-4} **B** 3.23×10^{-2} **C** 1.74×10^5 **D** 3.23×10^6